

Supervised Classification Model for Customer Churn Prediction

Objective

The objective of this project was to build and evaluate a supervised machine learning classification model capable of predicting customer churn. Customer churn refers to customers who discontinue a service or subscription. By predicting churn, businesses can identify at-risk customers and implement strategies to improve customer retention.

Dataset Description

A customer churn dataset containing 500 customer records was used for this project. The dataset included customer-related information such as age, monthly charges, tenure, contract type, internet service type, support calls, and churn status.

The target variable was Churn, where a value of 1 represented a customer who left the service and a value of 0 represented a customer who remained with the service.

Data Preprocessing

Several preprocessing steps were performed before model training. The dataset was loaded and checked for missing values. Categorical features, including ContractType and InternetService, were converted into numerical values using Label Encoding. The features and target variable were then separated, and the dataset was divided into training and testing sets using an 80:20 train-test split.

Models Used

Two supervised machine learning algorithms were implemented and compared:

1. Logistic Regression
2. Random Forest Classifier

Logistic Regression served as a baseline classification model, while Random Forest was used as an ensemble learning approach capable of handling more complex patterns within the data.

Model Evaluation

The models were evaluated using several performance metrics, including Accuracy, Precision, Recall, F1 Score, and ROC-AUC Score. In addition, 5-fold cross-validation was performed to assess model stability and generalization performance. A confusion matrix and ROC curve were also generated for visual evaluation.

Results

The Logistic Regression model achieved an accuracy of 0.77, a precision of 0.667, a recall of 0.160, an F1 score of 0.258, and a ROC-AUC score of 0.767. The average cross-validation accuracy for Logistic Regression was 0.734.

The Random Forest model achieved an accuracy of 0.73, a precision of 0.417, a recall of 0.200, an F1 score of 0.270, and a ROC-AUC score of 0.734. The average cross-validation accuracy for Random Forest was 0.716.

Discussion

Both models successfully classified customer churn using the provided dataset. Logistic Regression produced the highest overall accuracy and ROC-AUC score, indicating stronger predictive performance. The cross-validation results also showed that Logistic Regression generalized better across different data splits.

Although Random Forest achieved a slightly higher F1 score, its overall accuracy and ROC-AUC score were lower than those of Logistic Regression. Therefore, Logistic Regression demonstrated more reliable performance for this dataset.

Conclusion

This project successfully developed and evaluated supervised classification models for customer churn prediction. The workflow included data preprocessing, train-test splitting, cross-validation, model training, and performance evaluation.

Based on the obtained results, Logistic Regression outperformed Random Forest in terms of overall accuracy, ROC-AUC score, and cross-validation performance. Therefore, Logistic Regression was selected as the preferred model for predicting customer churn in this dataset. The project demonstrates how machine learning techniques can be applied to support customer retention strategies and data-driven decision-making.